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School of Informatics

Informatics Project Proposal

A Visual Trace Map of Edinburgh Place Names in Literature

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Abstract

This project investigates whether Edinburgh place names in literary texts can be visualised as an independent narrative structure rather than a rendering of geographic information. Drawing on the LitLong Edinburgh database — 548 literary works spanning 1687–2015, 2,135 distinct place names, and 48,056 mention records across 345 authors — the project will design a combined interactive visualisation system centred on two complementary research directions: per-author spatial fingerprints, which reveal how each author's place-name selection constitutes a distinct stylistic signature, and narrative topology networks, which expose the co-occurrence structure of places as defined purely by textual proximity. The system will be implemented using Python for data processing and D3.js for interactive visualisation, and will support comparative exploration of how different authors and literary traditions have imagined Edinburgh as a spatial entity.

Declaration of Own Work

I hereby declare that the work presented in this document is my own. Where ideas, text, or prior work from others have been drawn upon, they are appropriately cited. Use of generative AI tools has been acknowledged where applicable.

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1. Introduction

Place names in literary texts are conventionally treated as geographic references — data points to be projected onto a map. The LitLong Edinburgh project [1] demonstrated that large-scale text mining can surface thousands of Edinburgh place names across hundreds of literary works, and previous visualisations have rendered these mentions as interactive points and heatmaps on a geographic base layer [2]. This approach is informative, but it rests on an implicit assumption: that the meaning of a place name is determined by its location.

In literature, this assumption does not hold. When Robert Louis Stevenson writes of the narrow closes of the Old Town, the significance of that place lies not in its coordinates but in the narrative moment — which chapter it appears in, which other places surround it, and what that selection reveals about the author's imaginative geography. Two geographically adjacent places may never co-occur in a single text; two geographically distant places may be bound together in scene after scene. A geographical map cannot express this structure. As Loxley et al. [3] observe, the very act of text mining for place names foregrounds literature's selective construction of urban space: working-class neighbourhoods, industrial zones, and certain social spaces are persistently absent from the literary record, while others are endlessly revisited. This selectivity is itself a finding, but one that a geographic map obscures rather than reveals.

The project brief for this work explicitly calls for approaches that *move away from geographical maps* and instead explore abstract visualisation of sequence and co-occurrence [4]. This project takes that directive as a starting point for a deeper reframing: rather than treating place names as a geographic dataset to be rendered in a new visual form, it treats them as *narrative data* — structured by authorial choice, textual proximity, and literary tradition. The central question is: can place names in literary text be visualised as an independent narrative structure, one that reveals patterns invisible to any geographic rendering?

From this question, two complementary research directions emerge. The first asks whether the distribution of place names across an author's works constitutes a recognisable spatial signature — an abstract visual form that captures an author's distinctive imaginative geography. The second asks whether the co-occurrence of place names within and across texts reveals a narrative topology of the city, a relational map defined entirely by textual proximity rather than physical distance. Together, these two directions form the basis of a combined interactive visualisation system that will allow users to explore and compare how different authors and literary traditions have constructed Edinburgh as a spatial entity.

The project is grounded in digital humanities scholarship on geocriticism [5, 6] and the spatial analysis of literary corpora [7, 8], and builds directly on prior work in visualising fictional spatiality [9] and the broader field of text visualisation [10]. It extends these ideas to a richly structured, multi-author, multi-century Edinburgh corpus.

2. State of the Art

2.1. Literary Geography and Geocriticism

The relationship between literature and place has been a central concern of literary studies since at least the emergence of geocriticism as a formal methodology. Westphal's foundational work [5] frames geocriticism as the study of literary representations of space, understood as dynamic constructions rather than passive backdrops. Tally [6] broadens this to a general spatial literary studies encompassing any approach that attends to space, place, or mapping in literary texts. A key insight of this tradition is that literary place names are never neutral: they reflect the biases of authorship, genre, period, and cultural capital. As Loxley et al. [3] demonstrate in the context of Edinburgh, the pattern of which places are named and which are silenced encodes systematic social and aesthetic preferences that are invisible to individual close reading but become legible at the scale of a large corpus.

Franco Moretti's concept of *distant reading* [7] provides a complementary methodological frame. Moretti argues that literary history requires quantitative, aggregate analysis of large corpora rather than individual textual interpretation, and that abstract models — graphs, maps, trees — are the appropriate tools for this analysis. His work on mapping the geography of literary genres demonstrated that spatial analysis of large text collections can reveal structural patterns that no close reading of individual texts could surface. This project operates in the same tradition, applying it to the specific challenge of visualising place-name distributions in a curated Edinburgh literary corpus.

2.2. Text Mining and the LitLong Edinburgh Project

The Palimpsest project [1] developed the computational infrastructure that underpins the LitLong Edinburgh database. Using a pipeline combining named entity recognition (NER), coreference resolution, and georeferencing, the project extracted place name mentions from a large collection of Edinburgh-related literary texts and linked them to a custom Edinburgh gazetteer. The resulting database records, for each mention, the place name text, the source document, the author, the publication year, and the word-level position in the text. This level of granularity — including `sentence_id` and `page_id` fields — provides the relational structure necessary for co-occurrence analysis.

Gregory and Geddes [8] situate projects such as LitLong within the broader field of geohumanities, arguing that geographic information systems (GIS) and text mining together enable a new mode of spatial literary analysis at a scale previously impossible. However, they also note the limitations of conventional GIS representations for literary data: geographic maps privilege locational precision over narrative meaning, and render all place mentions as equivalent regardless of their narrative context or frequency. This project addresses that limitation directly.

2.3. Visualising Spatial Structures in Fiction

Stange and Dörk's *Novel City Maps* [9] is the most directly relevant prior work. Their system offers two views of a novel's spatial structure applied to three Berlin novels: a street map view, in which brightness encodes place-mention frequency to produce a spatial fingerprint of each novel, and a transit map (narrative) view, in which abstract metro-map aesthetics reveal the co-occurrence structures between important places within a single story. A key finding from their user evaluation is that the transit map metaphor, while visually striking, required explanation for users to interpret correctly — a design consideration directly relevant to the present project. Both views abandon strict geographic realism in favour of visual forms that foreground narrative relationships.

This project extends Stange and Dörk's approach in three significant ways. First, it applies the fingerprint and topology concepts to a *multi-author corpus* of 345 authors rather than individual novels, enabling cross-author comparison. Second, it separates the authorial fingerprint from the per-text topology, treating them as distinct but complementary analytical lenses. Third, it is driven by the richer relational structure available in the LitLong database, particularly the sentence-level and page-level co-occurrence information.

2.4. Graph and Network Visualisation

Network visualisation has been used extensively in digital humanities to reveal relational structures in literary and historical data. Character networks in novels have been studied as social graphs [11], revealing plot structure through patterns of character co-occurrence. The present project applies a closely analogous approach to place names: treating places as nodes and co-occurrence as edges, with the specific goal of comparing the resulting topological structure against geographic proximity.

Force-directed graph layout algorithms [12] are a standard approach for rendering such networks, but for this project the metro-map layout metaphor [13] is particularly appropriate: it prioritises connection structure over metric accuracy, precisely mirroring the shift from geographic to narrative space. The challenge of drawing schematic metro maps algorithmically — preserving octilinearity, avoiding crossings, and maintaining recognisable structure — is a well-studied problem in graph drawing that provides relevant technical context.

2.5. Information Visualisation for Literary Corpora

Jänicke et al.'s survey of visualisation for digital humanities [10] distinguishes between *close reading* tools (which support detailed engagement with individual texts) and *distant reading* tools (which support aggregate analysis of large corpora). The present project occupies an interesting position between these two modes: the fingerprint and topology views operate at the corpus level, but the interactive design allows users to drill down to individual authors and works, bridging aggregate pattern recognition with text-level exploration.

Hinrichs et al.'s work on speculative visualisation for literary collections [14] is also relevant: they argue that visualisation of literary data should not merely represent known structures but should create conditions for new interpretive questions to emerge. The abstract, non-geographic forms proposed here are designed in this spirit — not to answer a pre-formed question about Edinburgh's literary geography, but to make visible structures that readers and scholars had not previously been able to perceive.

2.6. Gap and Contribution

Existing tools for the LitLong Edinburgh dataset visualise place mentions geographically, preserving the assumption that locational meaning is primary. No prior work has applied fingerprint or topology-based visualisation to this specific corpus, nor has any prior work systematically compared author-level spatial signatures across a multi-century literary dataset. The contribution of this project is to design and implement visualisations grounded in narrative relationships rather than geographic ones, operating at the scale of a richly structured Edinburgh corpus, and supporting comparative exploration across authors and periods.

3. Implementation

3.1. Research Questions and Objectives

The project is oriented around two primary research questions:

1. **RQ1 (Spatial Fingerprints):** Does each author's distribution of place-name mentions constitute a visually distinctive spatial signature, and can abstract visualisation render

these signatures legible for comparative analysis?

2. **RQ2 (Narrative Topology):** Does the co-occurrence network of place names within and across texts reveal a narrative topology of Edinburgh that diverges systematically from its geographic topology, and can this divergence be made visible?

The primary objective is to design and implement a combined interactive system that addresses both questions. Secondary objectives include: (i) demonstrating the feasibility of narrative-structure visualisation at corpus scale; (ii) identifying at least one concrete pattern — an author whose spatial signature is visually distinctive, or a pair of places whose narrative proximity diverges significantly from their geographic proximity — that would not be discoverable from geographic maps alone.

3.2. Methodology and Approach

The project will produce a combined interactive visualisation system addressing the two research questions, developed in sequence.

Direction 1: Per-Author Spatial Fingerprints. The first direction operationalises RQ1. Each author in the corpus will be represented as a weighted frequency distribution over the set of 2,135 locations in `api_location.csv`, derived from their associated mention records in `api_locationmention.csv`. Because the corpus is highly imbalanced — Walter Scott alone accounts for a substantial fraction of the mentions — the representations will be normalised to enable meaningful cross-author comparison.

The visualisation will render each author's distribution as an abstract glyph: a radial or network-based form that encodes the structure of spatial attention without reference to geographic co-ordinates. The design space includes radar charts (where each axis corresponds to a frequently-mentioned place), hierarchical clustering visualisations (where places are grouped by co-occurrence affinity), and force-directed layouts restricted to each author's subgraph. Multiple candidate designs will be prototyped and evaluated against legibility criteria before the final form is selected. The design will support side-by-side comparison of author fingerprints, enabling a viewer to distinguish spatial languages at a glance. This extends the street-map fingerprint concept of Stange and Dörk [9] by abandoning geographic coordinates entirely.

Direction 2: Narrative Topology Networks. The second direction operationalises RQ2. The `sentence_id` and `page_id` fields in `api_locationmention.csv` allow direct computation of place co-occurrence at two granularities: within the same sentence (tight co-occurrence) and within the same page (loose co-occurrence). For each pair of places (p_i, p_j), the co-occurrence weight w_{ij} will be defined as the number of documents in which the two places co-occur within the chosen window. A weighted undirected graph $G = (V, E, w)$ will be constructed, where V is the set of places mentioned at least k times (a threshold to be determined empirically) and E contains all pairs with $w_{ij} > 0$.

This graph will be visualised using a force-directed layout [12] with a metro-map aesthetic [13], revealing topological clusters and connection patterns. A geographic overlay will be provided for reference, enabling direct visual comparison between narrative topology and geographic topology. The divergence between the two — places that are narratively proximate but geographically distant, and vice versa — is the primary analytical finding.

Combined Interface. The two directions will be integrated in a single interactive web application. Users will be able to: select an individual author or group of authors; switch between the fingerprint view (Direction 1) and the topology view (Direction 2); filter by time period or document collection; and hover over visual elements to surface underlying text passages. This design follows the close/distant reading continuum described by Jänicke et al. [10]: aggregate patterns are visible at the corpus level, but individual mentions remain accessible on demand.

Technical Stack. Data processing will be implemented in Python using `pandas` for data manipulation and `NetworkX` for graph construction and analysis. The interactive front-end will be implemented in JavaScript using D3.js [15], which provides the low-level rendering control required for bespoke abstract visualisation. The application will be deployed as a static browser-based tool with no server-side dependencies beyond file serving, ensuring reproducibility and portability.

3.3. Expected Outcomes

The primary deliverable is a functional, interactive web-based visualisation system supporting exploration of author spatial fingerprints and narrative topology networks across the LitLong Edinburgh corpus. The system will allow users to select authors or groups of authors, switch between views, filter by period or collection, and drill down to individual mentions. A written evaluation report and the full source code repository will accompany the implementation.

3.4. Evaluation Criteria

Evaluation will proceed along two dimensions. A **technical evaluation** will assess the correctness of the data processing pipeline (verified against known ground-truth cases, e.g. Scott's well-documented Edinburgh geography), the scalability of the co-occurrence computation across 48,056 records, and the interactive performance of the front-end application. A **design evaluation** will assess whether the visual forms are interpretable and generate meaningful analytical insights. This will be conducted through structured walkthroughs with at least five participants drawn from the project's target audience (students and researchers with an interest in digital humanities or literary geography), using a think-aloud protocol to capture interpretive responses.

Explicit success criteria are:

1. The system processes the full 48,056-record dataset without errors and produces correct co-occurrence counts verified against manual checks.
2. Fingerprints for the ten most prolific authors in the corpus are visually distinguishable from one another without reference to labels.
3. The narrative topology network for at least one selected author produces a connected graph with identifiable clusters, and at least one cluster can be interpreted in terms of known literary-geographic patterns.
4. At least four of five walkthrough participants report identifying a non-obvious spatial pattern (a surprising author fingerprint or an unexpected narrative proximity) that they would not have found using a geographic map.

3.5. Risk Assessment and Mitigations

3.6. Responsible Research Including Ethics

The dataset used in this project consists entirely of published literary texts and associated bibliographic metadata sourced from the LitLong Edinburgh database [1]. All source texts are either in the public domain or have been made available for academic research through the

Risk	**Likelihood**	**Mitigation**
Data quality issues (missing fields, inconsistent encoding, null coordinates)	Medium	Conduct exploratory data analysis in Week 1; implement explicit null handling; cross-validate against known authors (Scott, Stevenson)
Co-occurrence graph too sparse for meaningful topology at sentence level	Medium	Use page-level co-occurrence as fallback; apply frequency threshold k to restrict to well-attested locations
Corpus dominated by a small number of prolific authors (e.g. Walter Scott)	Medium	Apply per-author normalisation before fingerprint computation; provide both raw and normalised views
D3.js performance degradation with large graphs (>500 nodes)	Medium	Pre-aggregate and cache graph data; implement author/period filtering to reduce rendered node count
Fingerprint glyphs not visually distinguishable across authors	Medium	Prototype multiple glyph designs in Weeks 3–4; select design with highest discriminability in informal pilot
Time overrun on front-end development	Medium–High	Implement static visualisation first; add interactivity iteratively; treat combined interface as stretch goal

Table 1: Risk assessment and mitigations.

Palimpsest project's data agreements. No personal data about living individuals is collected or processed; all authors represented in the corpus are historical figures. No human participants are involved in the data collection or processing phases.

Evaluation walkthroughs will involve a small number of voluntary participants (students and colleagues) who will be informed of the project's purpose and the voluntary nature of their participation prior to the session. No identifying information will be recorded; walkthrough notes will record only task observations and verbal responses. This evaluation protocol does not require formal ethics review under the School of Informatics Ethics framework. If the evaluation is expanded to include a larger formal user study, ethics approval will be sought through the standard School process in advance.

Data management will comply with the University of Edinburgh's research data management guidelines. All code and processed data will be version-controlled in a private GitHub repository during development and made publicly available at dissertation submission. No data will be stored on personal devices beyond the project period.

4. Research Plan, Milestones, and Deliverables

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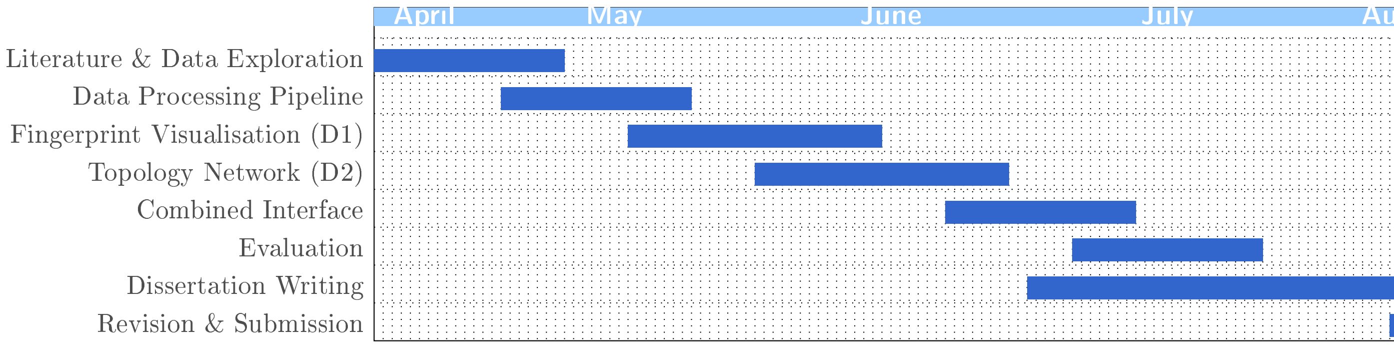

Figure 1: Gantt chart of project activities (April–August 2026).

Milestone	Target Date	Description
M_1	10 May	Literature review complete; data pipeline validated against ground truth
M_2	14 June	Author fingerprint visualisation functional (static prototype)
M_3	28 June	Narrative topology network visualisation functional
M_4	12 July	Combined interactive interface complete
M_5	26 July	Evaluation complete; dissertation draft begun
M_6	22 Aug	Dissertation submitted

Table 2: Project milestones.

References

[1] Beatrice Alex, Claire Grover, Jon Oberlander, Tara Thomson, Miranda Anderson, James Loxley, Uta Hinrichs, and Ke Zhou. Palimpsest: Improving assisted curation of loco-specific literature. *Digital Scholarship in the Humanities*, 32(suppl_1):i4–i16, 2017.

Deliverable	Target Date	Description
D_1	14 June	Author spatial fingerprint visualisation (static prototype)
D_2	28 June	Narrative topology network visualisation
D_3	12 July	Combined interactive web application
D_4	26 July	Evaluation report
D_5	22 Aug	Dissertation

Table 3: Project deliverables.

[2] LitLong Edinburgh. LitLong: Edinburgh [interactive map]. `http://litlong.org/`, 2015. Accessed 2026.

[3] James Loxley, Beatrice Alex, Miranda Anderson, Uta Hinrichs, Claire Grover, David Harris-Birtill, Tara Thomson, Aaron Quigley, and Jon Oberlander. 'Multiplicity embarrasses the eye': The digital mapping of literary Edinburgh. In *The Edinburgh Companion to Scottish Literature*. Edinburgh University Press, 2016.

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[5] Bertrand Westphal. *Geocriticism: Real and Fictional Spaces*. Palgrave Macmillan, 2011. Translated by Robert T. Tally Jr.

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[13] Martin Nöllenburg and Alexander Wolff. A mixed-integer program for drawing high-quality metro maps. In *Graph Drawing (GD 2005)*, volume 3843 of *Lecture Notes in Computer Science*, pages 321–333. Springer, 2006.

[14] Uta Hinrichs, Stefania Forlini, and Bridget Moynihan. Speculative practices: Utilizing InfoVis to explore untapped literary collections. *IEEE Transactions on Visualization and Computer Graphics*, 22(1):429–438, 2016.

[15] Michael Bostock, Vadim Ogievetsky, and Jeffrey Heer. D3: Data-driven documents. IEEE Transactions on Visualization and Computer Graphics, 17(12):2301–2309, 2011.

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